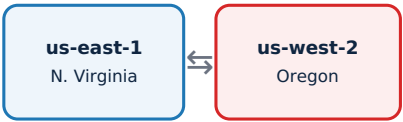


AWS Inter-Region Network Study

Latency, Jitter, Throughput & Retransmissions between
us-east-1 (N. Virginia) and us-west-2 (Oregon)



Method: iperf3, scheduled via cron, both directions
Forward samples (VA → OR): 308
Reverse samples (OR → VA): 191
Report generated by: analysis/generate_report.py

1. Project Overview

This project measures real-world network performance between two AWS EC2 instances in different geographic regions — **us-east-1 (N. Virginia)** and **us-west-2 (Oregon)** — using **iperf3**, run as a scheduled cron job on both machines, in **both directions**, over an extended period.

Testing the link in both directions matters because internet routing is not always symmetric — the path (and therefore the performance) from A→B can differ from B→A, even between the same two machines.

308 FORWARD SAMPLES VA → OR	191 REVERSE SAMPLES OR → VA	7 METRICS TRACKED PER SAMPLE	2 STATISTICAL TESTS PER METRIC
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What you get: two statistics CSVs (per-direction summary stats, and a forward-vs-reverse comparison with significance testing) plus 12 charts — 4 time-series per direction, and 4 side-by-side direction comparisons.

2. Metric Definitions

Every row in the raw data is one iperf3 test run. Here's exactly what each measured value means:

Metric	Definition
Average Latency ms	Average round-trip time for packets during the test. The core 'how fast does data get there and back' number. Lower is better.
Minimum Latency ms	Best (fastest) round-trip time seen during the test — the network's practical floor, close to the physical limit.
Maximum Latency ms	Worst (slowest) round-trip time seen during the test — reveals spikes and congestion moments hidden by the average.
Packet Loss %	Percentage of packets that never arrived. Even small loss (>1%) badly hurts TCP throughput and real-time apps like video/voice.
Jitter ms	Variation in latency between consecutive packets. Low jitter means a consistent, predictable network — important for streaming/VoIP even when average latency looks fine.
Throughput Mbps	Actual data transfer rate achieved during the test — the practical 'how much data can I move' number most people care about.
Retransmissions count	Number of TCP segments re-sent because they were lost or corrupted in transit. A proxy for underlying path quality, even when packet loss looks like 0% at the summary level.

3. Results

3.1 Summary Statistics

Forward direction — us-east-1 (Virginia) → us-west-2 (Oregon)

Metric	Mean	Median	Std Dev	Min	Max
Average Latency	56.62 ms	55.37 ms	2.62	54.71	68.42
Minimum Latency	56.24 ms	55.05 ms	2.56	54.60	67.27
Maximum Latency	62.17 ms	59.17 ms	8.49	54.80	106.34
Packet Loss	0.00 %	0.00 %	0.00	0.00	0.00
Jitter	1.30 ms	0.63 ms	1.72	0.02	11.19
Throughput	336.95 Mbps	317.50 Mbps	97.09	0.00	543.00
Retransmissions	657.93 count	604.00 count	495.75	0.00	4,199.00

Reverse direction — us-west-2 (Oregon) → us-east-1 (Virginia)

Metric	Mean	Median	Std Dev	Min	Max
Average Latency	56.41 ms	55.23 ms	1.97	54.66	62.54
Minimum Latency	56.03 ms	55.04 ms	1.84	54.61	61.85
Maximum Latency	61.48 ms	58.30 ms	9.12	54.72	105.19
Packet Loss	0.00 %	0.00 %	0.00	0.00	0.00
Jitter	1.22 ms	0.45 ms	1.95	0.02	10.87
Throughput	441.38 Mbps	473.00 Mbps	93.63	0.00	541.00
Retransmissions	458.82 count	127.00 count	698.46	0.00	3,061.00

3.2 Forward vs. Reverse — Is the Difference Real?

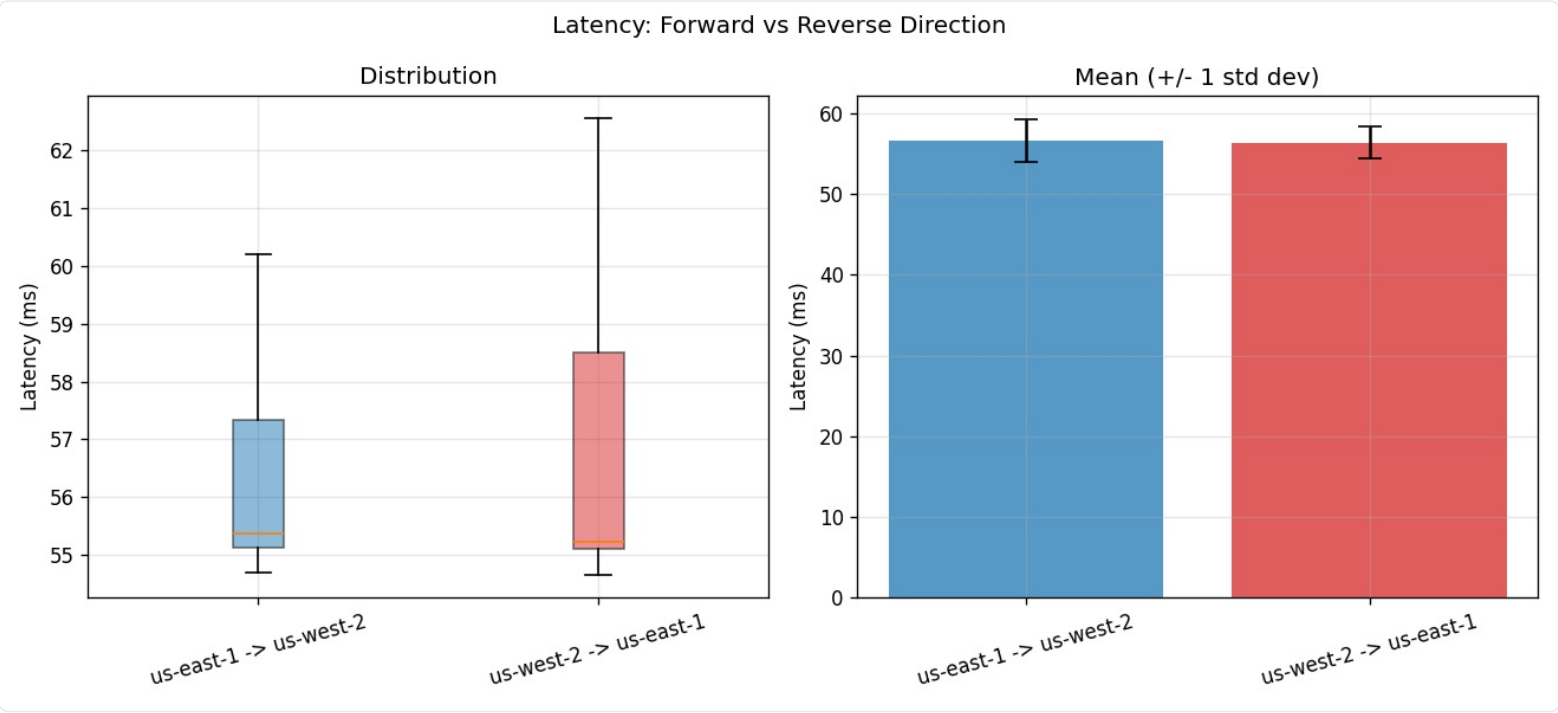
Each metric was compared between directions using Welch's t-test (does not assume equal variance) on the means. A result is flagged **Significant** when $p < 0.05$, meaning the observed difference is very unlikely to be due to random chance.

Metric	VA→OR Mean	OR→VA Mean	% Diff	P-Value	Result
Average Latency	56.62 ms	56.41 ms	+0.4%	0.295	Not significant
Minimum Latency	56.24 ms	56.03 ms	+0.4%	0.276	Not significant
Maximum Latency	62.17 ms	61.48 ms	+1.1%	0.401	Not significant
Packet Loss	0.00 %	0.00 %	—	—	Not significant
Jitter	1.30 ms	1.22 ms	+6.6%	0.639	Not significant
Throughput	336.95 Mbps	441.38 Mbps	-23.7%	1.93e-28	Significant
Retransmissions	657.93 count	458.82 count	+43.4%	6.64e-04	Significant

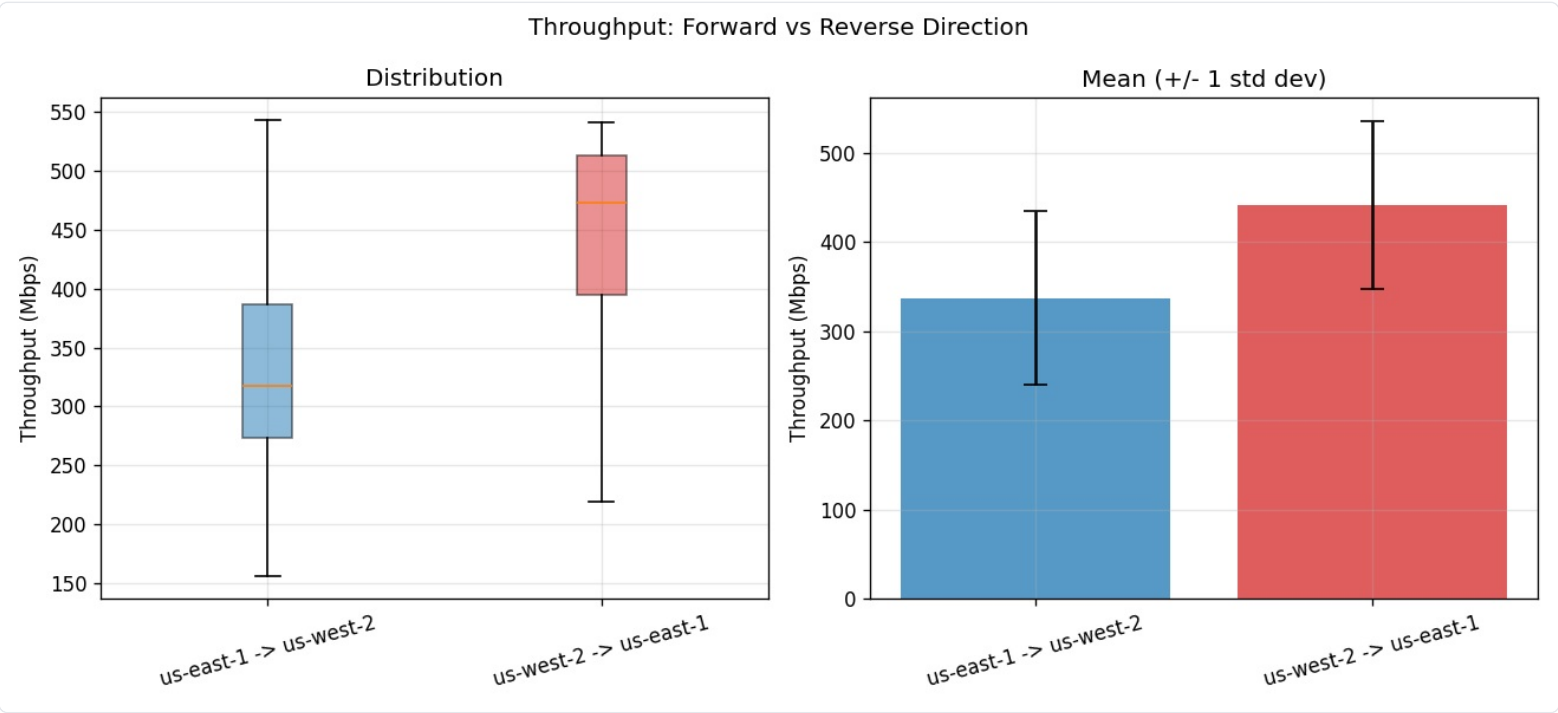
Key finding: Latency and jitter are statistically indistinguishable between directions — expected, since the physical route is roughly the same either way. Throughput and retransmissions, however, are **significantly asymmetric**: throughput running Oregon→Virginia is markedly higher, while Virginia→Oregon shows significantly more TCP retransmissions. That combination points to an asymmetric network path or a bottleneck on the sending side in Virginia, and would be worth investigating further with a traceroute/MTR comparison.

4. Direction Comparison Charts

Each chart shows the full distribution (box plot, left) and the mean $\pm$ 1 standard deviation (bar chart, right) for forward vs. reverse direction.

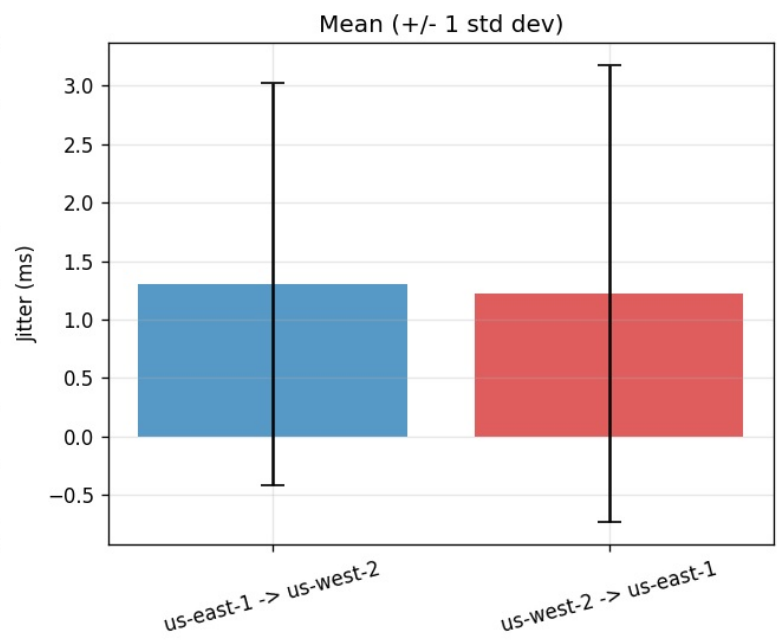
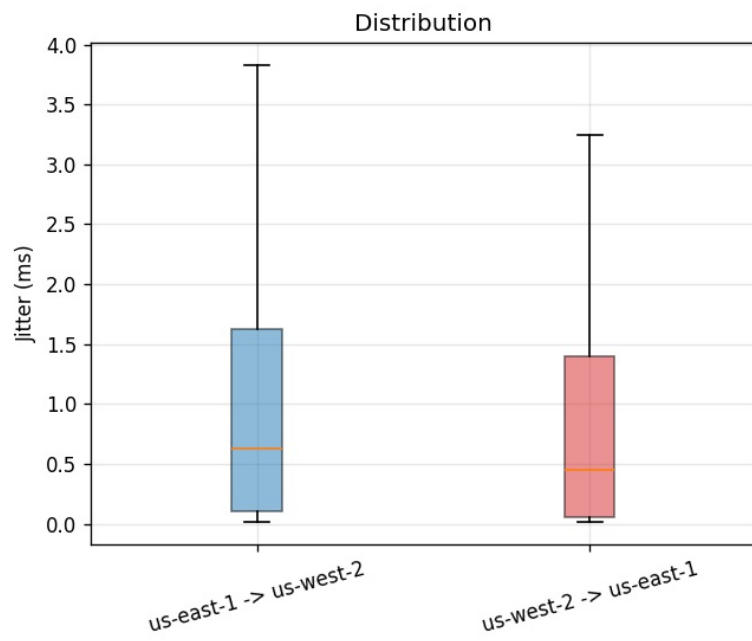


Latency: Forward vs. Reverse

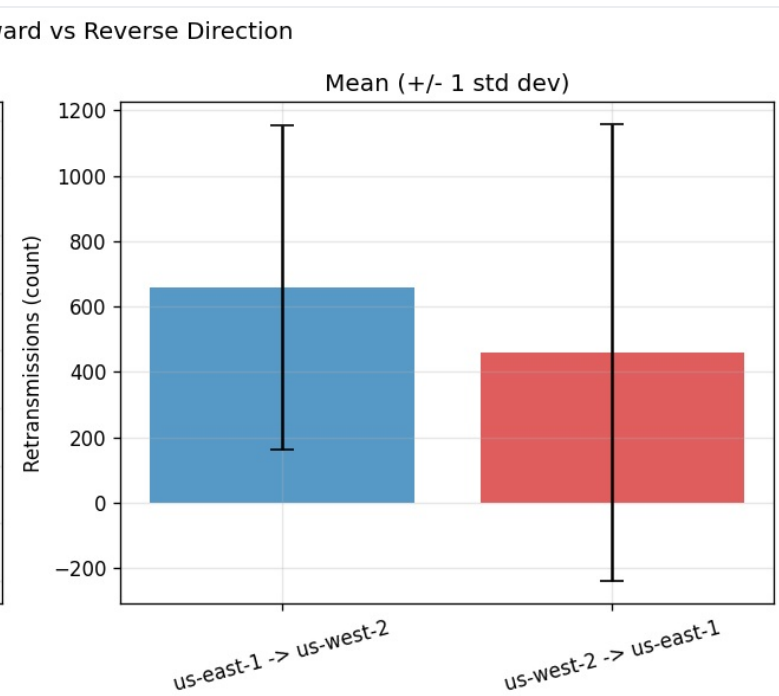
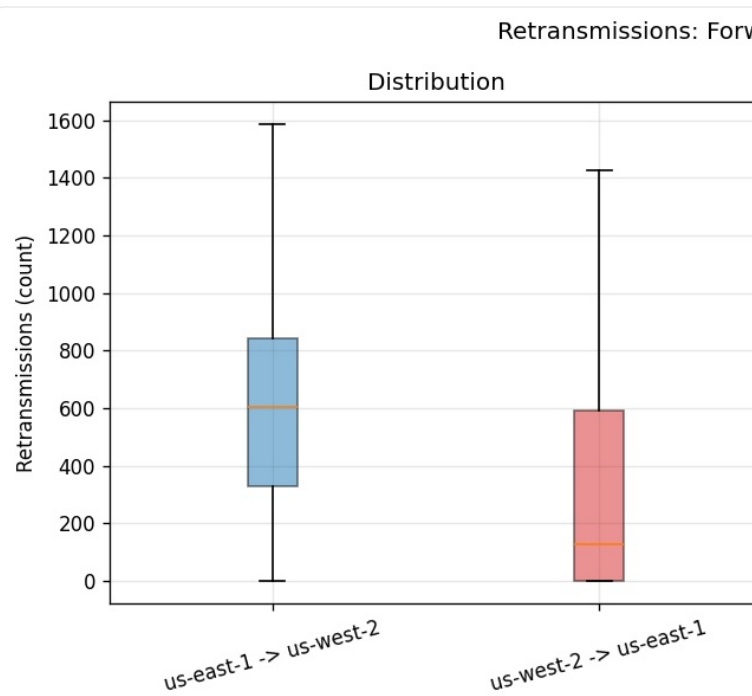


Throughput: Forward vs. Reverse

Jitter: Forward vs Reverse Direction



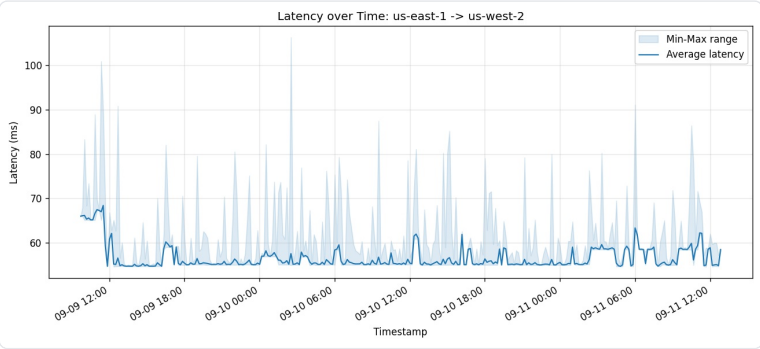
Jitter: Forward vs. Reverse



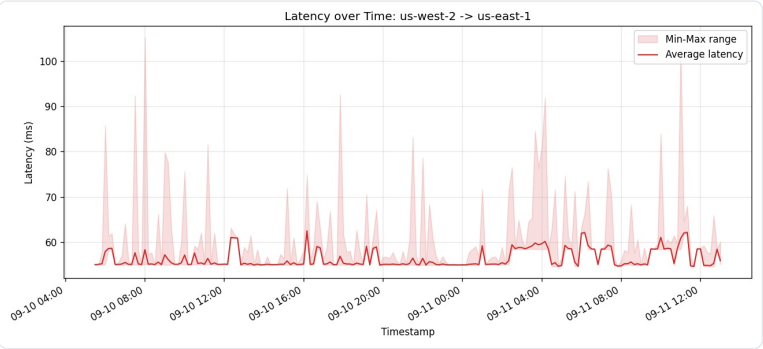
Retransmissions: Forward vs. Reverse

5. Time-Series Charts

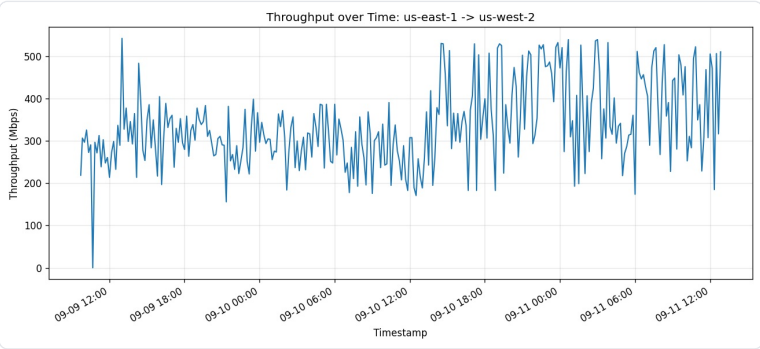
How each metric behaved over the full measurement window, for each direction.



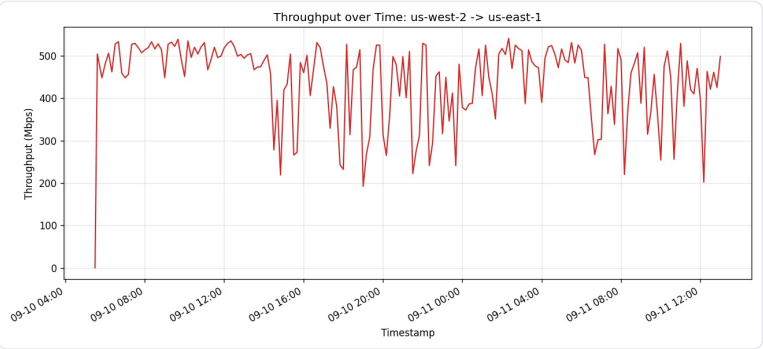
Latency over time — Virginia → Oregon



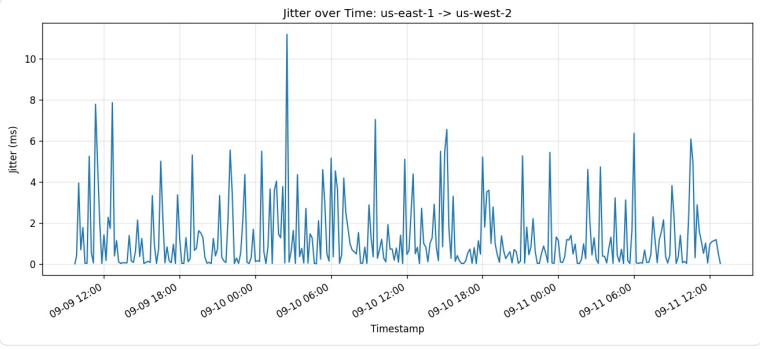
Latency over time — Oregon → Virginia



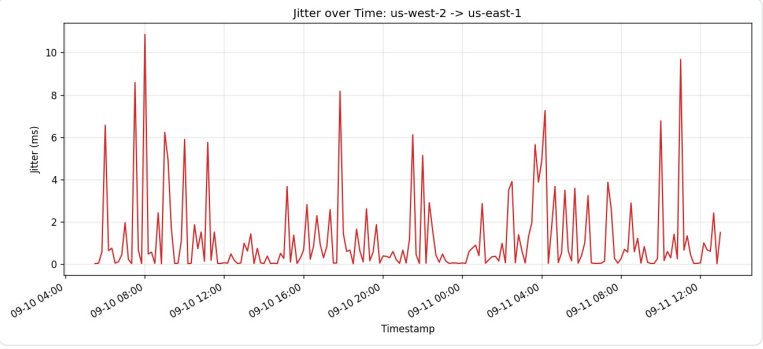
Throughput over time — Virginia → Oregon



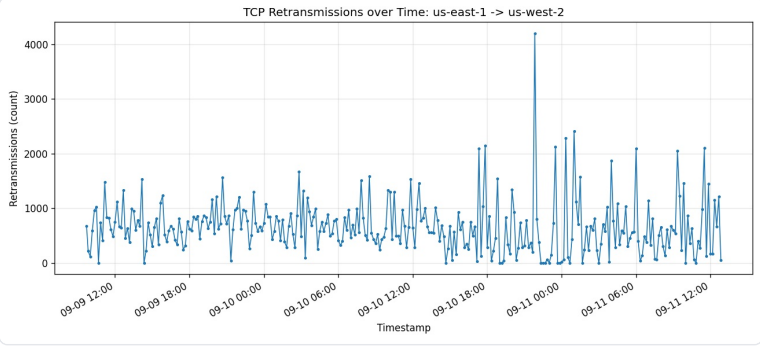
Throughput over time — Oregon → Virginia



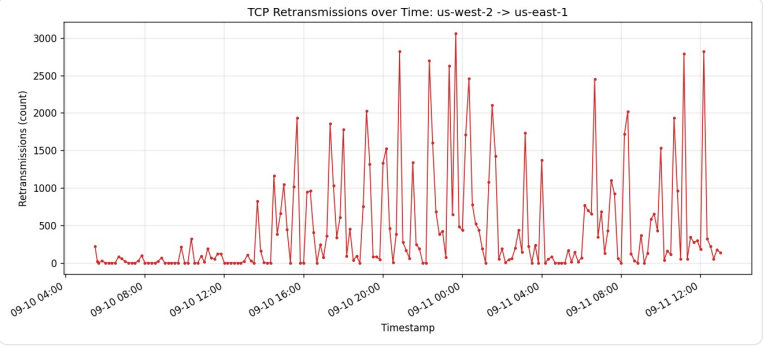
Jitter over time — Virginia → Oregon



Jitter over time — Oregon → Virginia



Retransmissions over time — Virginia → Oregon



Retransmissions over time — Oregon → Virginia

6. Summary

This study benchmarked bidirectional network performance between AWS us-east-1 (Virginia) and us-west-2 (Oregon) using 308 (east→west) and 191 (west→east) iperf3 measurements collected over several days. Round-trip latency and jitter were statistically indistinguishable between directions, consistent with a symmetric physical path. However, throughput was significantly asymmetric, and it was accompanied by significantly higher TCP retransmissions in the east→west direction. Packet loss was negligible (0%) in both directions. These results suggest the two directions of this inter-region link are not equivalent in practice, and that throughput-sensitive applications may see measurably better performance sending data from Oregon to Virginia than the reverse.